magical it feels. Magic makes the development experience easier, but it can be much more difficult to reason about in a scaled application. For example, hooks and automatic relationship loading significantly increase debugging time, but they also decrease initial development effort.

Ecto makes a point of reducing magic. Operations are explicit and can be easily reasoned about when reading an application's source code. This philosophy has resulted in Ecto not having hooks or automatic relationship loading, in addition to other small changes that are present throughout the library. Many people love that Ecto gets rid of magic.

In general, Ecto feels more like SQL, but it still protects you when it comes to security and performance best practices. ActiveRecord feels more like accessing an object. This is great in some situations, like querying a system via the Rails console, but it results in more magic that can be difficult to reason about.

Ecto Concepts

There are three main roles that Ecto satisfies: querying the database, translating data from the database into application structures, and providing ways to change data. Here's how Ecto tackles each of these roles:

Query the Database

Ecto has a built-in query language called Ecto.Query[6] that you use to write queries in Elixir. These queries are securely translated into SQL, and the database executes the query. Ecto makes heavy use of macros, so queries are validated at compile time for security and correctness.

Ecto's Query language isn't SQL, but it feels very similar. It's rare that you are unable to write a query that you want to write. Ecto.Query also lets you load and reference relationships using built-in functions.

Translate Data from the Database

Ecto schemas[7] map data from your database into an application struct using type definitions provided in a schema. Schemas also define associations between other schemas, and Ecto provides functions to work with these associations.

Schemas don't provide any access to the database, so you don't have to worry about accidental database queries once a schema is realized from

6. https://hexdocs.pm/ecto/Ecto.Query.html
7. https://hexdocs.pm/ecto/Ecto.Schema.html